

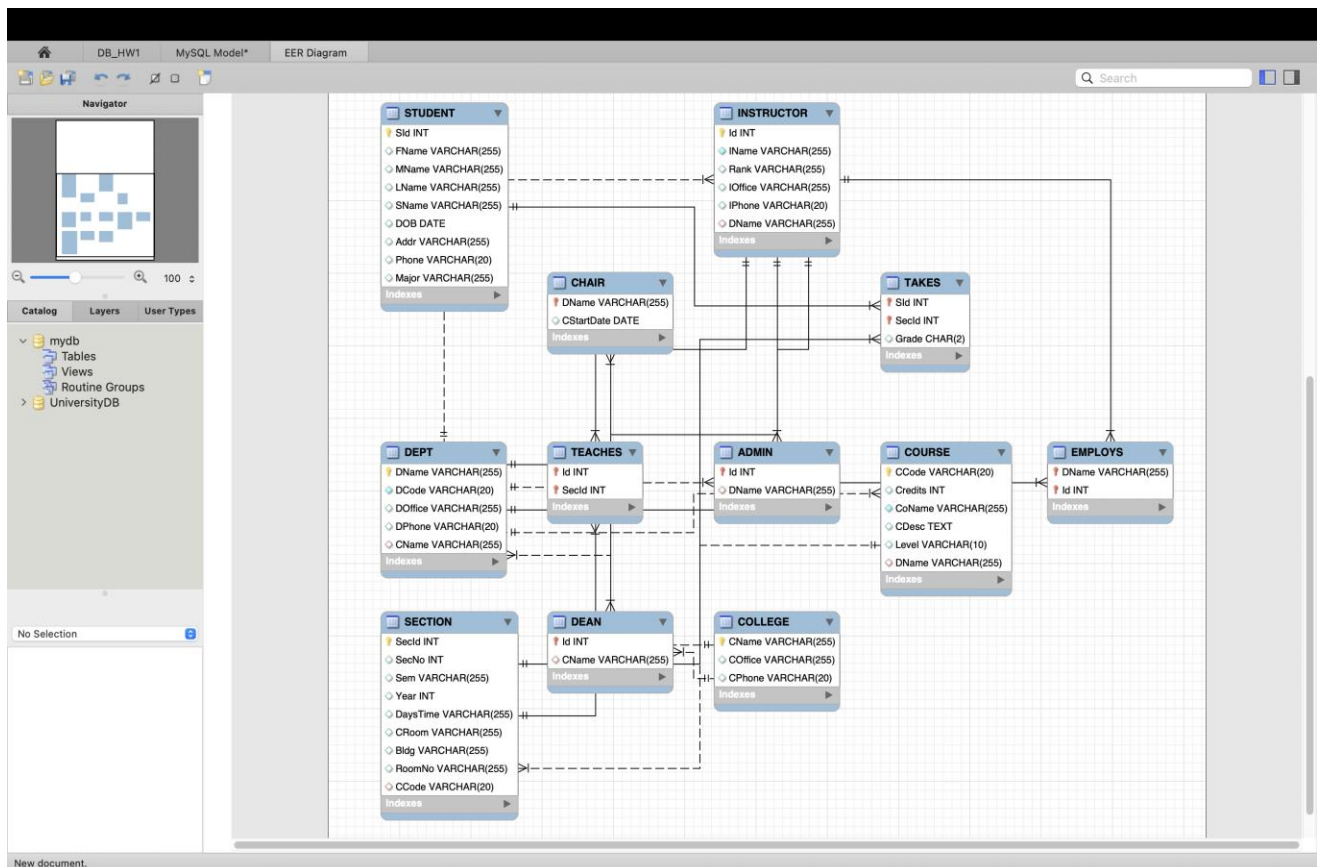
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CIS 552: Database Design

Homework – 1 Part - 2:

1. Building an ER Schema Using SQL developer Data model.

The below image shows all the entities of ERWIN diagram which are listed in the question, Like STUDENT, INSTRUCTOR, SECTION, COURSE etc.



The relationship of them mentioned in the ER diagram as follows:

- **COLLEGE employs DEAN:** (1,1) to (0,1)
- **COLLEGE employs ADMIN:** (0, N) to (1,1)
- **DEPT has CHAIR:** (0, N) to (1,1)
- **DEPT employs INSTRUCTOR:** (0, N) to (0,1)
- **DEPT offers COURSE:** (1,1) to (0, N)
- **COURSE has SECTIONS:** (0, N) to (1,1)
- **INSTRUCTOR teaches SECTION:** (0, N) to (1,1)
- **STUDENT takes SECTION:** (0, N) to (5, N)

The below images show the SQL code demonstration of the diagram shown above.

The screenshot shows a database management tool interface with the following components:

- Top Bar:** Includes a home icon, a tab labeled "DB_HW1", and various utility icons.
- Left Panel (SCHEMAS):** A tree view showing the database structure:
 - sys (selected)
 - UniversityDB
 - Tables
 - Views
 - Stored Procedures
 - Functions
- Bottom Left:** "Object Info" and "Session" tabs, with "Schema: sys" selected.
- Central Query Editor:** Contains SQL code for creating a database and tables:


```

1  -- Create the database if it doesn't already exist
2  • CREATE DATABASE IF NOT EXISTS UniversityDB;
3
4  -- Use the newly created or existing database
5  • USE UniversityDB;
6
7  -- Create the COLLEGE table
8  • CREATE TABLE COLLEGE (
9      CName VARCHAR(255) PRIMARY KEY,
10     COffice VARCHAR(255),
11     CPhone VARCHAR(20)
12 );
13
14 -- Create the DEPT (DEPARTMENT) table
15 • CREATE TABLE DEPT (
16     DName VARCHAR(255) PRIMARY KEY,
17     DCode VARCHAR(20) NOT NULL UNIQUE,
18     DOffice VARCHAR(255),
19     DPhone VARCHAR(20),
20     CName VARCHAR(255),
21     FOREIGN KEY (CName) REFERENCES COLLEGE(CName)
22 );
23
24 -- Create the CHAIR table
25 • CREATE TABLE CHAIR (
26     DName VARCHAR(255),
27     CStartDate DATE,
28     PRIMARY KEY (DName),
29     FOREIGN KEY (DName) REFERENCES DEPT(DName)
30 );
      
```
- Right Panel:** "Context Help" and "Snippets" tabs. The "Context Help" tab is active, displaying a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."
- Action Output:** A table showing the execution results of the queries:

	Time	Action	Response	Duration / Fetch Time
11	14:45:59	CREATE TABLE TEACHERS (Id INT, SecId INT, PRIMARY KEY (Id, SecId), FOREIGN...	0 row(s) affected	0.0030 sec
12	14:45:59	CREATE TABLE TAKES (Sid INT, SecId INT, Grade CHAR(2), PRIMARY KEY (Sid, S...	0 row(s) affected	0.0032 sec
13	14:45:59	CREATE TABLE ADMIN (Id INT PRIMARY KEY, DName VARCHAR(255), FOREIGN KE...	0 row(s) affected	0.0035 sec
14	14:45:59	CREATE TABLE DEAN (Id INT PRIMARY KEY, CName VARCHAR(255), FOREIGN KE...	0 row(s) affected	0.0030 sec
15	14:45:59	CREATE TABLE EMPLOYS (DName VARCHAR(255), Id INT, PRIMARY KEY (DName,...	0 row(s) affected	0.0035 sec
- Bottom Status Bar:** Displays "Query Completed".

As shown above in the code, we have created a database called UniversityDB and deployed all the code in that database.

After that we have created code for entities like COLLEGE, DEPT etc...

DB_HW1

Administration Schemas Query 1

Limit to 50000 rows

SCHEMAS

Filter objects

sys

UniversityDB

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: sys

```

23 -- Create the CHAIR table
24
25 CREATE TABLE CHAIR (
26     DName VARCHAR(255),
27     CStartDate DATE,
28     PRIMARY KEY (DName),
29     FOREIGN KEY (DName) REFERENCES DEPT(DName)
30 );
31
32 -- Create the INSTRUCTOR table
33
34 CREATE TABLE INSTRUCTOR (
35     Id INT PRIMARY KEY AUTO_INCREMENT,
36     IName VARCHAR(255) NOT NULL,
37     Rank VARCHAR(255),
38     IOffice VARCHAR(255),
39     IPHONE VARCHAR(20),
40     DName VARCHAR(255),
41     FOREIGN KEY (DName) REFERENCES DEPT(DName)
42 );
43
44 -- Create the STUDENT table
45
46 CREATE TABLE STUDENT (
47     SId INT PRIMARY KEY AUTO_INCREMENT,
48     FName VARCHAR(255),
49     MName VARCHAR(255),
50     LName VARCHAR(255),
51     SName VARCHAR(255),
52     DOB DATE,
53     Addr VARCHAR(255),
54     Phone VARCHAR(20),
55     Major VARCHAR(255)
56 );
57
58 -- Create the COURSE table
59
60 CREATE TABLE COURSE (
61     CCode VARCHAR(20) PRIMARY KEY,
62     Credits INT,
63     CoName VARCHAR(255) NOT NULL,
64     CDesc TEXT,
65     Level VARCHAR(10),
66     DName VARCHAR(255),
67     FOREIGN KEY (DName) REFERENCES DEPT(DName)
68 );
69
70 -- Create the SECTION table
71
72 CREATE TABLE SECTION (
73     SecId INT PRIMARY KEY AUTO_INCREMENT,
74     SecNo INT,
75     Sem VARCHAR(255),
76     Year INT,
77     DaysTime VARCHAR(255),
78     CRoom VARCHAR(255),
79     Bldg VARCHAR(255),
80     RoomNo VARCHAR(255),
81     CCode VARCHAR(20),
82     FOREIGN KEY (CCode) REFERENCES COURSE(CCode)
83 );

```

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Action	Time	Action	Response	Duration / Fetch Time
11	14:45:59	CREATE TABLE TEACHES (Id INT, SecId INT, PRIMARY KEY (Id, SecId), FOREIGN KEY (Id) REFERENCES INSTRUCTOR(Id), FOREIGN KEY (SecId) REFERENCES SECTION(SecId))	0 row(s) affected	0.0036 sec
12	14:45:59	CREATE TABLE TAKES (SId INT, SecId INT, Grade CHAR(2), PRIMARY KEY (SId, SecId), FOREIGN KEY (SId) REFERENCES STUDENT(SId), FOREIGN KEY (SecId) REFERENCES SECTION(SecId))	0 row(s) affected	0.0032 sec
13	14:45:59	CREATE TABLE ADMIN (Id INT PRIMARY KEY, DName VARCHAR(255), FOREIGN KEY (DName) REFERENCES DEPT(DName))	0 row(s) affected	0.0035 sec
14	14:45:59	CREATE TABLE DEAN (Id INT PRIMARY KEY, CName VARCHAR(255), FOREIGN KEY (CName) REFERENCES COURSE(CCode))	0 row(s) affected	0.0030 sec
15	14:45:59	CREATE TABLE EMPLOYEES (DName VARCHAR(255), Id INT, PRIMARY KEY (DName, Id), FOREIGN KEY (DName) REFERENCES DEPT(DName))	0 row(s) affected	0.0035 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

DB_HW1

Administration Schemas Query 1

Limit to 50000 rows

SCHEMAS

Filter objects

sys

UniversityDB

Tables

Views

Stored Procedures

Functions

Object Info Session

Schema: sys

```

48 DOB DATE,
49 Addr VARCHAR(255),
50 Phone VARCHAR(20),
51 Major VARCHAR(255)
52 );
53
54 -- Create the COURSE table
55
56 CREATE TABLE COURSE (
57     CCode VARCHAR(20) PRIMARY KEY,
58     Credits INT,
59     CoName VARCHAR(255) NOT NULL,
60     CDesc TEXT,
61     Level VARCHAR(10),
62     DName VARCHAR(255),
63     FOREIGN KEY (DName) REFERENCES DEPT(DName)
64 );
65
66 -- Create the SECTION table
67
68 CREATE TABLE SECTION (
69     SecId INT PRIMARY KEY AUTO_INCREMENT,
70     SecNo INT,
71     Sem VARCHAR(255),
72     Year INT,
73     DaysTime VARCHAR(255),
74     CRoom VARCHAR(255),
75     Bldg VARCHAR(255),
76     RoomNo VARCHAR(255),
77     CCode VARCHAR(20),
78     FOREIGN KEY (CCode) REFERENCES COURSE(CCode)
79 );
80
81 -- Create the EMPLOYEES table
82
83 CREATE TABLE EMPLOYEES (
84     DName VARCHAR(255),
85     Id INT,
86     PRIMARY KEY (DName, Id),
87     FOREIGN KEY (DName) REFERENCES DEPT(DName)
88 );
89
90 -- Create the STUDENT table
91
92 CREATE TABLE STUDENT (
93     SId INT PRIMARY KEY AUTO_INCREMENT,
94     FName VARCHAR(255),
95     MName VARCHAR(255),
96     LName VARCHAR(255),
97     SName VARCHAR(255),
98     DOB DATE,
99     Addr VARCHAR(255),
100     Phone VARCHAR(20),
101     Major VARCHAR(255)
102 );
103
104 -- Create the INSTRUCTOR table
105
106 CREATE TABLE INSTRUCTOR (
107     Id INT PRIMARY KEY AUTO_INCREMENT,
108     IName VARCHAR(255) NOT NULL,
109     Rank VARCHAR(255),
110     IOffice VARCHAR(255),
111     IPHONE VARCHAR(20),
112     DName VARCHAR(255),
113     FOREIGN KEY (DName) REFERENCES DEPT(DName)
114 );
115
116 -- Create the CHAIR table
117
118 CREATE TABLE CHAIR (
119     DName VARCHAR(255),
120     CStartDate DATE,
121     PRIMARY KEY (DName),
122     FOREIGN KEY (DName) REFERENCES DEPT(DName)
123 );
124
125 -- Create the TAKES table
126
127 CREATE TABLE TAKES (
128     SId INT,
129     SecId INT,
130     Grade CHAR(2),
131     PRIMARY KEY (SId, SecId),
132     FOREIGN KEY (SId) REFERENCES STUDENT(SId),
133     FOREIGN KEY (SecId) REFERENCES SECTION(SecId)
134 );
135
136 -- Create the TEACHES table
137
138 CREATE TABLE TEACHES (
139     Id INT,
140     SecId INT,
141     PRIMARY KEY (Id, SecId),
142     FOREIGN KEY (Id) REFERENCES INSTRUCTOR(Id),
143     FOREIGN KEY (SecId) REFERENCES SECTION(SecId)
144 );
145
146 -- Create the ADMIN table
147
148 CREATE TABLE ADMIN (
149     Id INT PRIMARY KEY,
150     DName VARCHAR(255),
151     FOREIGN KEY (DName) REFERENCES DEPT(DName)
152 );
143
144 -- Create the DEAN table
145
146 CREATE TABLE DEAN (
147     Id INT PRIMARY KEY,
148     CName VARCHAR(255),
149     FOREIGN KEY (CName) REFERENCES COURSE(CCode)
150 );
151
152 -- Create the EMPLOYEES table
153
154 CREATE TABLE EMPLOYEES (
155     DName VARCHAR(255),
156     Id INT,
157     PRIMARY KEY (DName, Id),
158     FOREIGN KEY (DName) REFERENCES DEPT(DName)
159 );

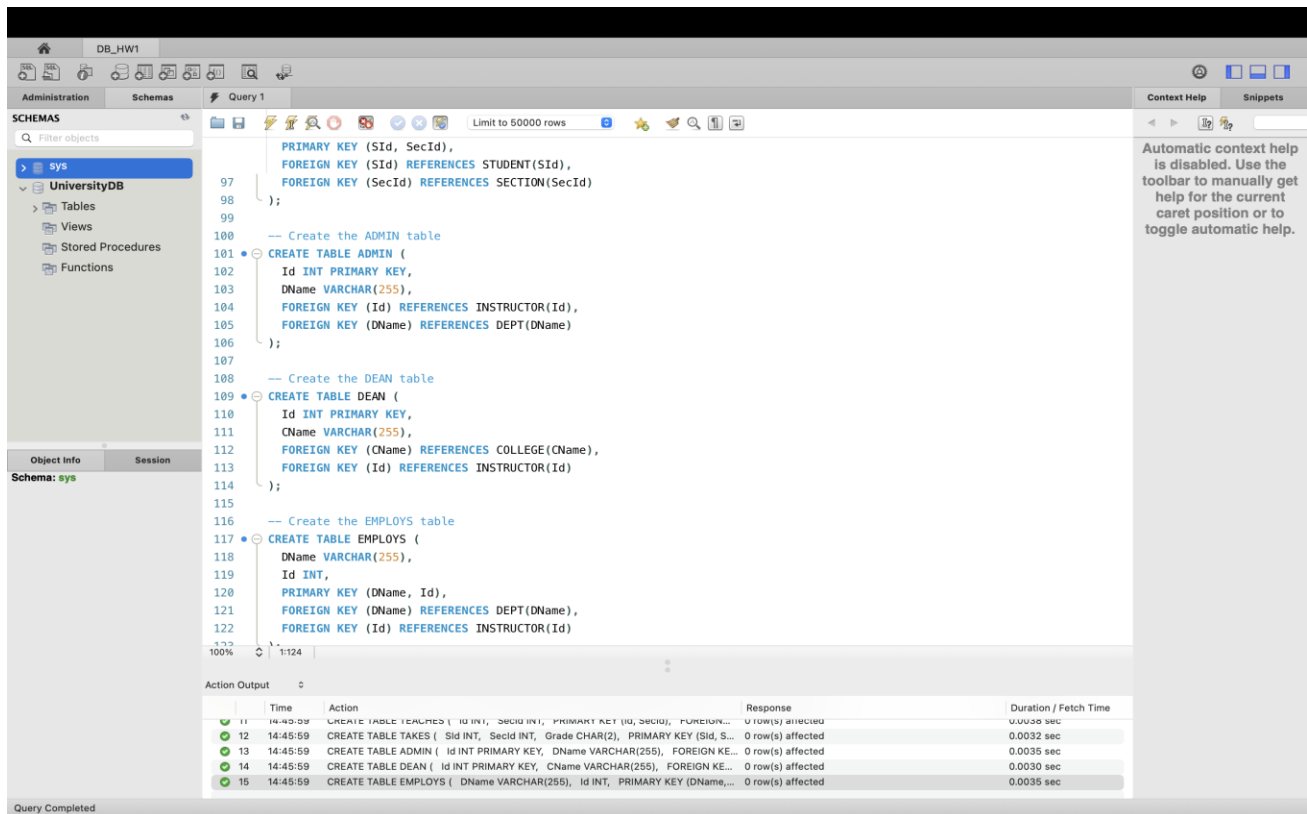
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Query Completed

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After this code has been run successfully, by using reverse engineering we will plot the diagram in SQL workbench.

The plot and SQL code of this one had been attached to you.

Conclusion:

From this what I have learnt is that as follows:

- Writing my first ever SQL code
- Downloading SQL workbench
- Connecting it to Database by using SQL server.
- Creating an EER diagram by using SQL code.
- Adding the entities and making connections in the Database.
- Using Reverse engineering for plotting EER diagram.

- Downloading the EER plot and SQL codes.

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